AGILE LAB FAT EXAM

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23MIS0422

Payment Processing System

Problem Statement: Ensure payments are processed securely and correctly.

Tasks:

1. Develop payment validation logic

2. Write JUnit tests for transaction success/failure

3. Implement CI/CD pipeline with automated testing

> Set up version control using GitHub

> Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits Automate build process using Maven > Integrate automated unit tests in pipeline Configure CD pipeline for automatic deployment to staging/production

4. Deploy services on Docker and Kubernetes

> Containerize application using Docker (write Dockerfile)

> Build and push Docker image to container registry (Docker Hub)

Create Kubernetes deployment and service YAML files

EXPLORER

OPEN EDITORS

GROUP 1

pom.xml

GROUP 2

PaymentService.java 1

PAYMENT-SYSTEM

target

PaymentService.java 1

pom.xml

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.payment</groupId>
    <artifactId>payment-system</artifactId>
    <version>1.0</version>

    <build>
        <plugins>
            <plugin>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.10.1</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

PaymentService.java 1

PaymentService.java > Language Support for Java(TM) by Red Hat > PaymentService

```java
public class PaymentService {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) throws Exception {
        System.out.println("Payment System Started...");

        while (true) {
            Thread.sleep(5000);
            System.out.println("Processing Payments...");
        }
    }
}
```

EXPLORER

OPEN EDITORS

pom.xml

PAYMENT-SYSTEM

target

PaymentService.java 1

pom.xml

OUTLINE

TIMELINE

PROJECTS

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

```
PS C:\Users\prave\OneDrive\Desktop\payment-system> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.payment:payment-system >-------------------
[INFO] Building payment-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ payment-system ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\prave\OneDrive\Desktop\payment-system\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ payment-system ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ payment-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\prave\OneDrive\Desktop\payment-system\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ payment-system ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ payment-system ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ payment-system ---
[WARNING] JAR will be empty - no content was marked for inclusion!
[INFO] Building jar: C:\Users\prave\OneDrive\Desktop\payment-system\target\payment-system-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.076 s
[INFO] Finished at: 2026-04-13T10:21:34+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\prave\OneDrive\Desktop\payment-system>
```

https://github.com/Praveen272006/payment-system

Praveen272006 / payment-system

Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

payment-system Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file Add file Code

Praveen272006 Initial Payment CI/CD commit 1c91bd7 · now 1 Commit

target	Initial Payment CI/CD commit	now
Dockerfile	Initial Payment CI/CD commit	now
PaymentService.java	Initial Payment CI/CD commit	now
pom.xml	Initial Payment CI/CD commit	now

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

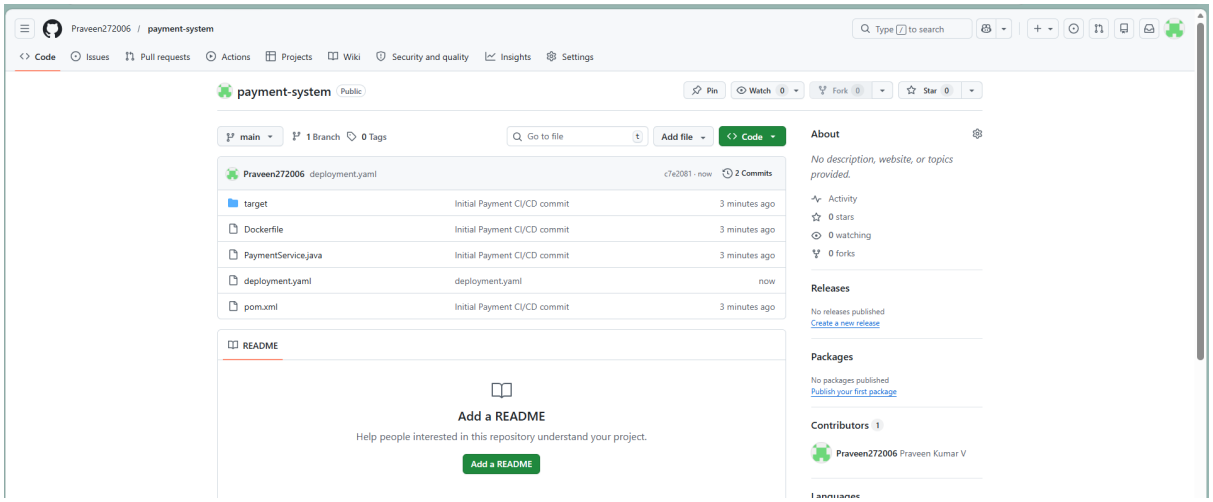
Publish your first package

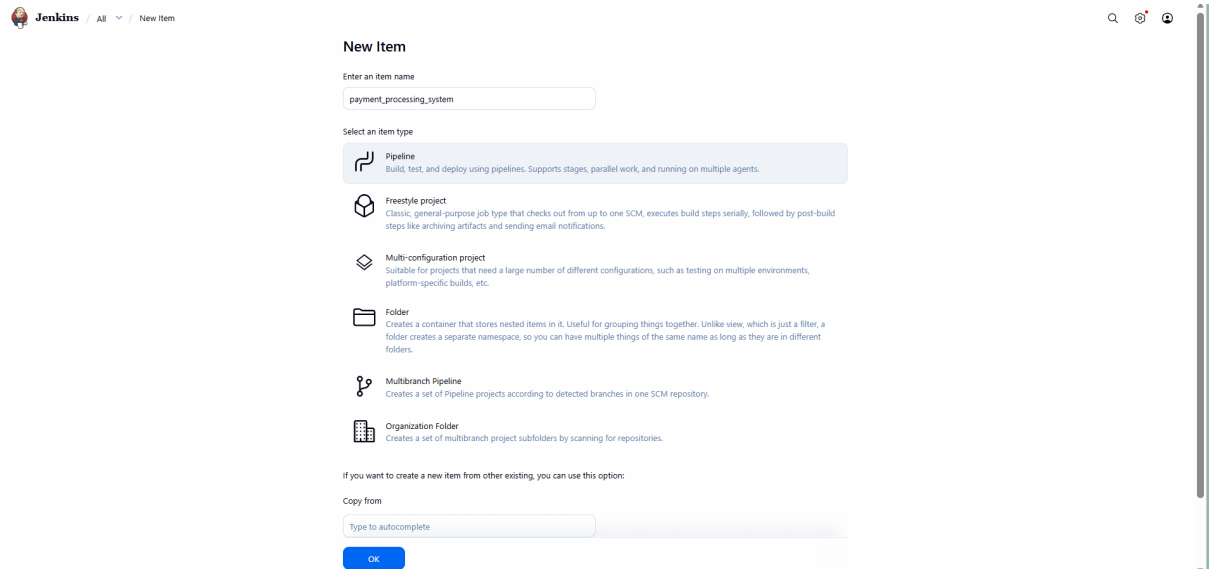
Contributors 1

Praveen272006 Praveen Kumar V

Languages

Java 70.9% Dockerfile 29.1%


Praveen272006 / payment-system
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
payment-system
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Praveen272006 deployment.yaml
c7e2081 · now
2 Commits
target
Initial Payment CI/CD commit
3 minutes ago
Dockerfile
Initial Payment CI/CD commit
3 minutes ago
PaymentService.java
Initial Payment CI/CD commit
3 minutes ago
deployment.yaml
deployment.yaml
now
pom.xml
Initial Payment CI/CD commit
3 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Praveen272006 Praveen Kumar V


Jenkins / All / New Item
New Item
Enter an item name
payment_processing_system
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
If you want to create a new item from other existing, you can use this option:
Copy from
Type to autocomplete
OK

Jenkins / payment_processing_syst... / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
pipeline {

    agent any

    environment {
        DOCKER_IMAGE = "praveen272006/java-app:latest"
    }

    stages {

        stage('Clone Repo') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/Praveen272006/payment-system.git'
            }
```

☑ Use Groovy Sandbox ?

Pipeline Syntax

Advanced

Advanced

Save Apply

Jenkins / payment_processing_syst... / #1

```
d38f7db82f14: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
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3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
3024ca44526d: Waiting
d38f7db82f14: Pushed
3024ca44526d: Pushed
latest: digest: sha256:d6320ee44cc888e35c185e8e049f255dfd59b7deb0284273dfd88636074e7384 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```
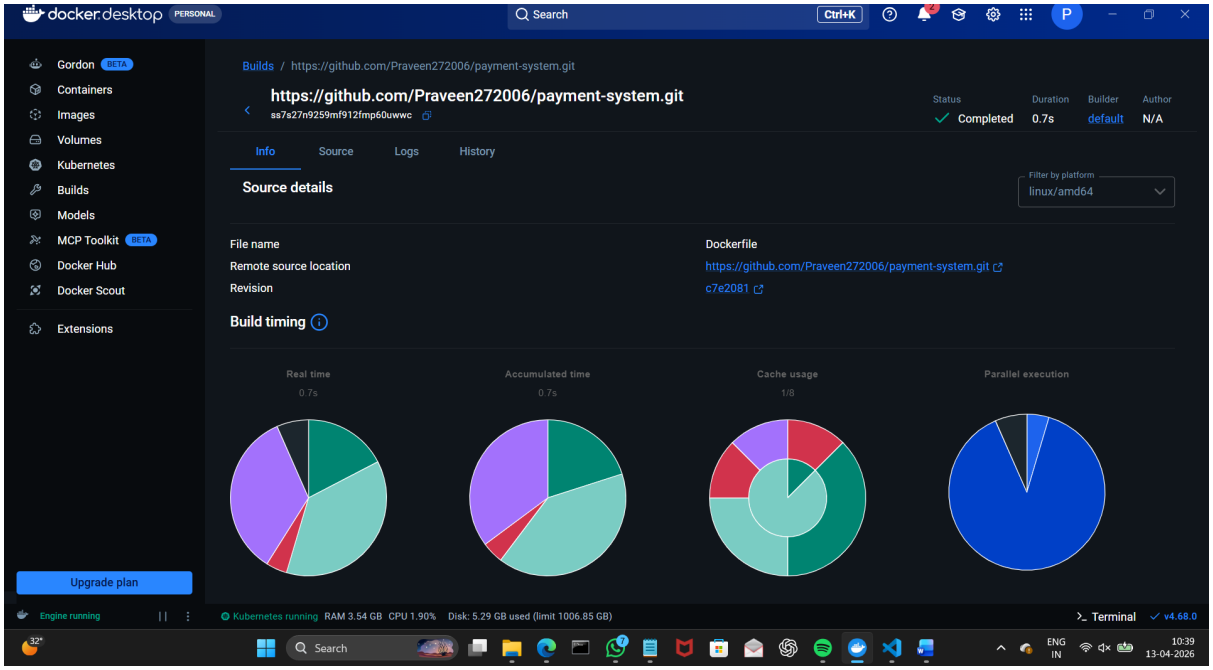
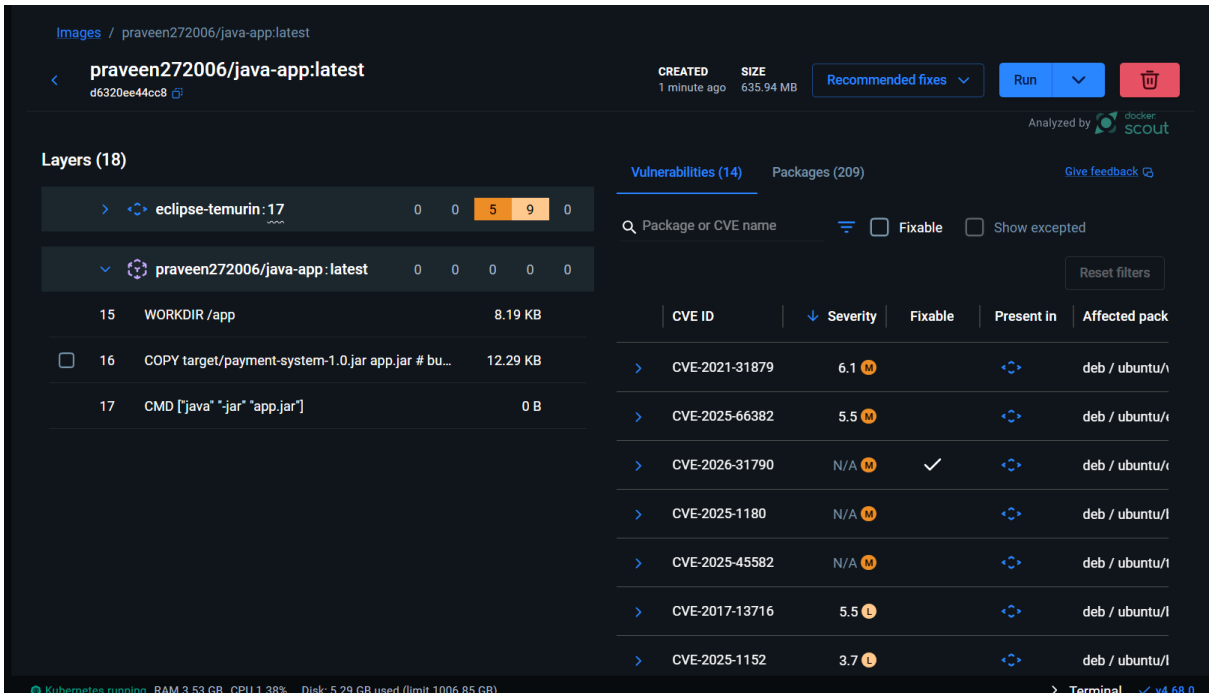


docker desktop PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Builds / https://github.com/Praveen272006/payment-system.git
https://github.com/Praveen272006/payment-system.git
ss7s27n9259mf912fmp60uwwc
Status
Completed
Duration
0.7s
Builder
default
Author
N/A
Info
Source
Logs
History
Source details
Filter by platform
linux/amd64
File name
Dockerfile
Remote source location
https://github.com/Praveen272006/payment-system.git
Revision
c7e2081
Build timing
Real time
0.7s
Accumulated time
0.7s
Cache usage
1/8
Parallel execution
Engine running
Kubernetes running RAM 3.54 GB CPU 1.90% Disk: 5.29 GB used (limit 1006.85 GB)
Terminal
v4.68.0
Search
ENG
IN
10:39
13-04-2026


Images / praveen272006/java-app:latest
praveen272006/java-app:latest
d6320ee44cc8
CREATED
1 minute ago
SIZE
635.94 MB
Recommended fixes
Run
Analyzed by docker scout
Layers (18)
eclipse-temurin:17
0 0 5 9 0
praveen272006/java-app:latest
0 0 0 0 0
15 WORKDIR /app 8.19 KB
16 COPY target/payment-system-1.0.jar app.jar # bu... 12.29 KB
17 CMD ["java" "-jar" "app.jar"] 0 B
Vulnerabilities (14)
Packages (209)
Give feedback
Package or CVE name
Fixable
Show excepted
Reset filters
CVE ID
Severity
Fixable
Present in
Affected pack
CVE-2021-31879 6.1 M deb / ubuntu/
CVE-2025-66382 5.5 M deb / ubuntu/
CVE-2026-31790 N/A M deb / ubuntu/
CVE-2025-1180 N/A M deb / ubuntu/
CVE-2025-45582 N/A M deb / ubuntu/
CVE-2017-13716 5.5 L deb / ubuntu/
CVE-2025-1152 3.7 L deb / ubuntu/

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\prave\OneDrive\Desktop\payment-system> kubectl get nodes
NAME                   STATUS   ROLES           AGE   VERSION
desktop-control-plane  Ready    control-plane   13h   v1.34.3
PS C:\Users\prave\OneDrive\Desktop\payment-system> kubectl get svc
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   13h
PS C:\Users\prave\OneDrive\Desktop\payment-system>
```

Java: Ready | Praveen272006 (now) | Ln 22, Col 15 | Spaces: 2 | UTF-8 | CRLF | YAML | Finish Setup | Go Live

Search

ENG IN 10:38 13-04-2026